Long LI

PhD in Applied Mathematics
EU resident card (2024-2034)

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Research Interests

Stochastic Modeling, Uncertainty Quantification, Data Assimilation, Ensemble Forecasting, Numerical Methods, Scientific Computing, Geophysical Fluid Dynamics, Turbulence

Education

2017 – 2020 **PhD in Applied Mathematics**, *Université de Rennes I*, France

2015 – 2017 **Master in Applied Mathematics**, *Université Grenoble Alpes*, France

2012 – 2015 **Bachelor in Mathematics**, *Université Jean Monnet Saint Etienne*, France

Research Experience

10/2025 – **CNRS Postdoctoral Researcher**, *Institut de Physique de Nice*, France
Scalar turbulence driven by multifractal random flows, advised by Simon Thalabard
Keywords: stochastic physics, intermittency, Gaussian multiplicative chaos

04/2021 – 07/2025 **INRIA Starting Research Positions**, *Centre Inria de l'Université de Rennes*, France

- Projects Participation:
  - Research Associate in STUOD (European Research Council Project)
  - Postdoctoral Fellow in MEDIATION (French Priority Research Program)
- Scientific Contributions:
  - Stochastic, data-driven parameterizations of subgrid-scale ocean processes
  - GPU-accelerated, auto-differentiable solvers with efficient stochastic integrators
  - Stochastic modeling of air–sea interactions and boundary layer turbulence

2017 – 2020 **Doctoral Researcher**, *Institut de recherche mathématique de Rennes*, France

- PhD Thesis: *Stochastic modeling and numerical simulation of ocean dynamics* [HAL], supervised by Etienne Mémin and Werner Bauer
  - Developed a hierarchy of stochastic large-scale ocean models
  - Investigated uncertainty quantification and ensemble forecasting

2017 **Research Intern**, *Laboratoire Jean Kuntzmann*, Grenoble, France
6 months

- MSc Thesis: *Global sensitivity analysis for a parametrized diffusion process*, supervised by Clémentine Prieur and Pierre Étoré
  - Developed a novel uncertainty quantification method for stochastic differential equations by integrating Feynman-Kac formulae with polynomial chaos expansion

2016 **Research Intern**, *Laboratoire Jean Kuntzmann*, Grenoble, France
3 months *Numerical methods for stochastic differential equations*, supervised by Clémentine Prieur

Supervision and Teaching

2021 – 2024, 3 years — **PhD Co-supervisor:** Francesco Tucciarone (Computer Science, University of Rennes I), *Stochastic parametrization of ocean models through high-resolution observations* [Thesis]. Publication of three book chapters and a journal article.

2024, 4 months — **Master Intern Co-supervisor:** Antoine Guines (Mathematical Engineering, INSA Rouen), *Numerical study of a stochastic ocean model in the Mediterranean configuration*. A detailed comparison between deterministic and stochastic simulations.

2013 – 2015, 30 hours — **Tutor for undergraduate students:** *Exercise sessions in Analysis and Algebra*, Faculty of Sciences and Techniques, Université Jean Monnet.

Peer Reviewing

Journals — Journal of Advances in Modeling Earth Systems (JAMES), Geoscientific Model Development (GMD), Frontiers in Marine Science (Physical Oceanography)

Books — Chapters in Mathematics of Planet Earth (MPE)

Languages

Chinese — Native

English — Fluent (C1)

French — Fluent (C1)

Computer Skills

Programming — Python, Fortran, C/C++, PyTorch, Julia, Matlab, R

HPC — OpenMP, MPI, CUDA

Tools — Vim, Jupyter Notebook, Markdown, LaTeX, ParaView, NetCDF, XIOS

Systems — Linux, macOS, Git, SVN, Docker, OAR, SLURM, shell/batch scripting

Publications

Journal Articles

[doi] [pdf] [code] Li, L., Mémin, E., Chapron, B. (2025). A generalized stochastic formulation of the Ekman-Stokes model with statistical analyses. Journal of Physical Oceanography, 55, 1389–1407.

[doi] [pdf] [code] Tucciarone, F., Li, L., Mémin, E., Chandramouli, P. (2025). Derivation and numerical assessment of a stochastic large-scale hydrostatic primitive model. Journal of Advances in Modeling Earth Systems, 17, e2024MS004783.

[doi] [pdf] [code] Clement, S., Blayo, E., Debreu, L., Brankart, J.-M., Brasseur, P., Li, L., Mémin, E. (2025). Link between stochastic grid perturbation and location uncertainty framework. Journal of Advances in Modeling Earth Systems, 17, e2024MS004528.

[doi] [pdf] [code] Thiry, L., Li, L., Mémin, E., Roullet, G. (2024). A unified formulation of quasi-geostrophic and shallow water equations via projection. Journal of Advances in Modeling Earth Systems, 16, e2024MS004510.

[doi] [pdf] [code] Thiry, L., Li, L., Roullet, G., Mémin, E. (2024). MQGeometry-1.0: a multi-layer quasi-geostrophic solver on non-rectangular geometries. Geoscientific Model Development, 17, 1749–1764.

[doi] [pdf] [code] Li, L., Deremble, B., Lahaye, N., Mémin, E. (2023). Stochastic data-driven parameterization of unresolved eddy effects in a baroclinic quasi-geostrophic model. Journal of Advances in Modeling Earth Systems, 15, e2022MS003297.

[doi] [pdf] [code] Brecht, R., Li, L., Bauer, W., Mémin, E. (2021). Rotating shallow water flow under location uncertainty with a structure-preserving discretization. Journal of Advances in Modeling Earth Systems, 13 (12), e2021MS002492.

[doi] [pdf] Bauer, W., Chandramouli, P., Chapron, B., Li, L., Mémin, E. (2020). Deciphering the role of small-scale inhomogeneity on geophysical flow structuration: a stochastic approach. Journal of Physical Oceanography 50, 983-1003.

[doi] [pdf] Bauer, W., Chandramouli, P., Li, L., Mémin, E. (2020). Stochastic representation of mesoscale eddy effects in coarse-resolution barotropic models. Ocean Modelling 151, 101646.

[doi] [pdf] [code] Resseguier, V., Li, L., Jouan, G., Derian, P., Mémin, E., Chapron, B. (2020). New trends in ensemble forecast strategy: uncertainty quantification for coarse-grid computational fluid dynamics. Archives of Computational Methods in Engineering, 1886-1784.

[doi] [pdf] Étoré, P., Prieur, C., Pham, D., Li, L. (2020). Global sensitivity analysis for models described by stochastic differential equations. Methodology and Computing in Applied Probability 22, 803-831.

Book Chapters

[doi] [pdf] [code] Li, L., Mémin, E., Chapron, B. (2026). A stochastic Ekman-Stokes model for coupled ocean-atmosphere-wave dynamics. Stochastic Transport in Upper Ocean Dynamics IV, Mathematics of Planet Earth, vol 15. Springer, 105-131.

[doi] [pdf] Tucciarone, F., Li, L., Mémin, E., Thiry, L. (2025). Transport noise defined from wavelet transform for model-based stochastic ocean models. Stochastic Transport in Upper Ocean Dynamics III, Mathematics of Planet Earth, vol 13. Springer, 287-303.

[doi] [pdf] [code] Mémin, E., Li, L., Lahaye, N., Tissot, G., and Chapron, B. (2024). Linear wave solutions of a stochastic shallow water model. Stochastic Transport in Upper Ocean Dynamics II, Mathematics of Planet Earth, vol 11. Springer, 223–245.

[doi] [pdf] Tucciarone, F., Mémin, E., Li, L. (2024). Data driven stochastic primitive equations with dynamic modes decomposition. Stochastic Transport in Upper Ocean Dynamics II, Mathematics of Planet Earth, vol 11. Springer, 321–336.

[doi] [pdf] Jamet, Q., Mémin, E., Dumas, F., Li, L., Garreau, P. (2024). Toward a stochastic parameterization for oceanic deep convection. Stochastic Transport in Upper Ocean Dynamics II, Mathematics of Planet Earth, vol 11. Springer, 143–157.

[doi] [pdf] [code] Li, L., Mémin, E., Tissot, G. (2023). Stochastic parameterization with dynamic mode decomposition. Stochastic Transport in Upper Ocean Dynamics, Mathematics of Planet Earth, vol 10. Springer, 179–193.

[doi] [pdf] [code] Thiry, L., Li, L., Mémin, E. (2023). Modified (hyper-)viscosity for coarse-resolution ocean models. In Stochastic Transport in Upper Ocean Dynamics, 273-285. Springer, Cham.

[doi] [pdf] Tucciarone, F., Mémin, E., Li, L. (2023). Primitive equations under location uncertainty: analytical description and model development. Stochastic Transport in Upper Ocean Dynamics, Mathematics of Planet Earth, vol 10. Springer, 287–300.

[doi] [pdf] Fiorini, C., Boulvard, P.-M., Li, L., Mémin, E. (2023). A Two-step numerical scheme in time for surface quasi-geostrophic equations under location uncertainty. Stochastic Transport in Upper Ocean Dynamics, Mathematics of Planet Earth, vol 10. Springer, 57–67.

Presentations

[slides] Li, L. (Sep. 2025). Stochastic approaches to air–sea boundary layer dynamics. 6th STUOD Annual Workshop, ICMS The Bayes Centre, Edinburgh, UK.

[slides] Li, L. (Mar. 2025). Stochastic modeling in geophysical fluid dynamics. Invited Seminar, ANGE Project-Team, INRIA, Paris, France.

[slides] Li, L. (Oct. 2024). A stochastic formulation of the Ekman-Stokes model. Atelier sur la représentation des fines échelles océaniques dans les simulations numériques, Ifremer, Plouzané, France.

[slides] Li, L. (Oct. 2024). Stochastic modeling in geophysical fluid dynamics. Invited Seminar, CALISTO Project-Team, Inria, Valbonne, France.

[slides] Li, L. (Sep. 2024). A generalized stochastic formulation of the Ekman-Stokes model with statistical analyses. 5th STUOD Annual Workshop, Inria, Rennes, France.

[slides] Li, L. (Sep. 2024). Stochastic modeling in geophysical fluid dynamics. Invited Seminar, Research Center for Mathematics and Interdisciplinary Sciences, Shandong University, Qingdao, China.

[slides] Li, L. (Sep. 2023). Uncertainty quantification of mesoscale air-sea interaction under transport noise. 4th STUOD Annual Workshop, Ifremer, Plouzané, France.

[slides] Li, L. (Mar. 2023). Linear wave solutions of a stochastic shallow water model. 22nd STUOD Sandbox Workshop, Online.

[slides] Li, L. (Nov. 2022). Stochastic transport in an idealized ocean-atmosphere coupled model. AIRSEA Team Seminar, Inria, Grenoble, France.

[slides] Li, L. (Sep. 2022). Stochastic transport in an idealized ocean-atmosphere coupled model. 3rd STUOD Annual Workshop, Imperial College London, London, UK.

[slides] Li, L. (May 2022). Stochastic transport in an idealized ocean-atmosphere coupled system. EGU General Assembly, Vienna, Austria.

[slides] Li, L. (Nov. 2021). Numerically modelling Location Uncertainty in geophysical fluid dynamics. 11th STUOD Sandbox Workshop, Online.

[slides] Li, L. (Sep. 2021). Statistically data-driven modelling location uncertainty in mesoscale dynamics. 2nd STUOD Annual Workshop, Online.

[slides] Li, L. (Apr. 2021). Stochastic modeling of mesoscale eddies. EGU General Assembly, Online.

[slides] Li, L. (Feb. 2021). Stochastic modeling of the oceanic mesoscale eddies. 6th STUOD Sandbox Workshop, Online.

[slides] Li, L. (Sep. 2020). Stochastic modeling of the oceanic mesoscale eddies. 1st STUOD Annual Workshop, Online.

[slides] Li, L. (Sep. 2020). Rotating shallow water flow under location uncertainty. 3rd STUOD Sandbox Workshop, Online.

[slides] Li, L. (Sep. 2019). Stochastic modeling of mesoscale eddies in oceanic dynamics. Workshop on Frontiers of Uncertainty Quantification in Fluid Dynamics, Pisa, Italy.

[poster] Li, L. (Jul. 2019). Stochastic modeling of mesoscale eddies in barotropic wind-driven circulation. Workshop on Stochastic Parameterizations and Their Use in Data Assimilation, London, UK.

[slides] Li, L. (Apr. 2019). Oceanic dynamics under Location Uncertainty - Towards a consistent stochastic modeling. Workshop Conservation Principles, Data and Uncertainty in Atmosphere-Ocean Modeling, Potsdam, Germany.